

A Decade of Federal Tax Data: What Statistical Analysis Reveals About U.S. Tax Policy

Wonseok Lee

Virginia Tech × Fullstack Academy DA Bootcamp

What Is TCJA?

Tax Cuts and Jobs Act

Signed: December 2017

Effective: Tax Year 2018

Largest U.S. tax overhaul since 1986

Question: Did it work as expected?

Key Changes

- Reduced individual income tax brackets
- Doubled standard deduction
- Capped state and local tax (SALT) deduction
- Lowered corporate tax rate

How I Investigated



Data Source

IRS Statistics of Income — Historic Table 2

11 years: 2012–2022

6,468 records | 50 states + DC | 11 income
brackets



Tools & Methods

Python (pandas, scipy, sklearn)

Pearson correlation, ANOVA, t-tests

Pre/Post comparative analysis (t-tests)

Tableau for interactive visualization

6,468

Records

51

States + DC

11

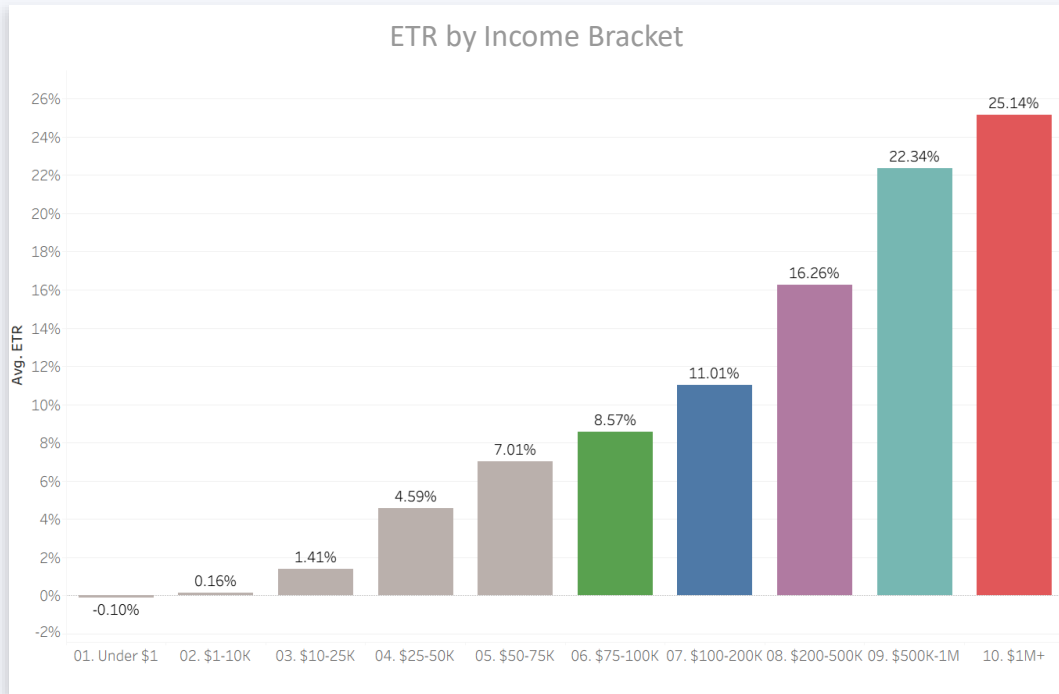
Years

11

Brackets

Is the U.S. Tax System Progressive?

Answer: Yes



Pearson r

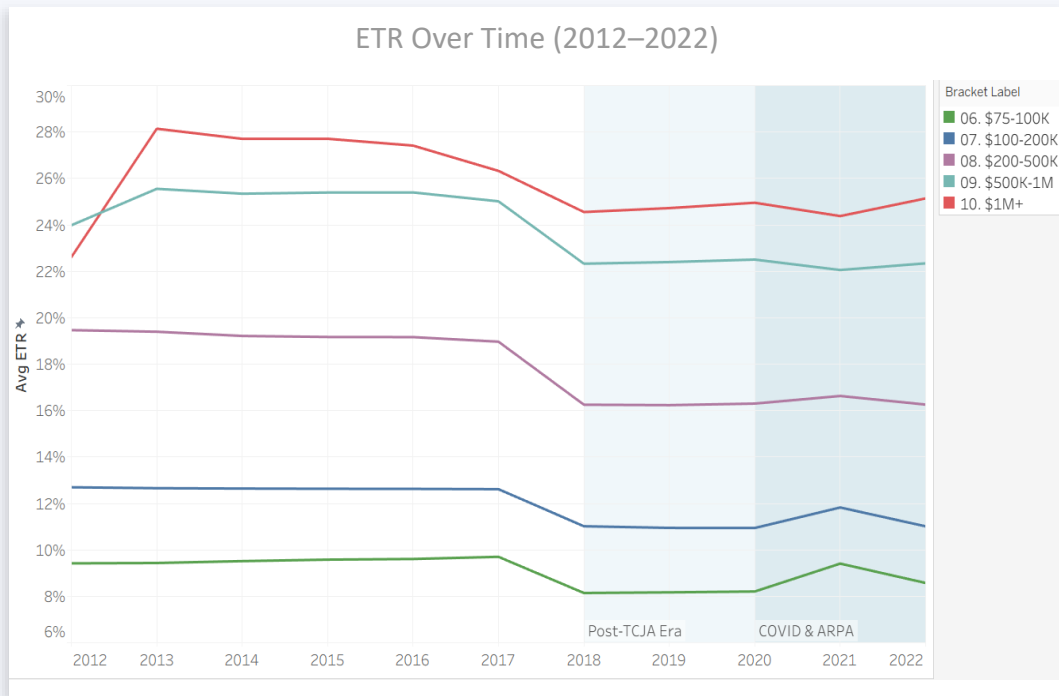
0.9602

$p < 0.001$ ✓ SIGNIFICANT

Effective Tax Rate by Income

Under \$10K	0.2%
\$50–75K	7.0%
\$200–500K	16.3%
\$1M+	25.1%

Did TCJA Reduce Effective Tax Rates?



Pre-TCJA (2012–2017)

10.85%

Post-TCJA (2018–2022)

9.54%

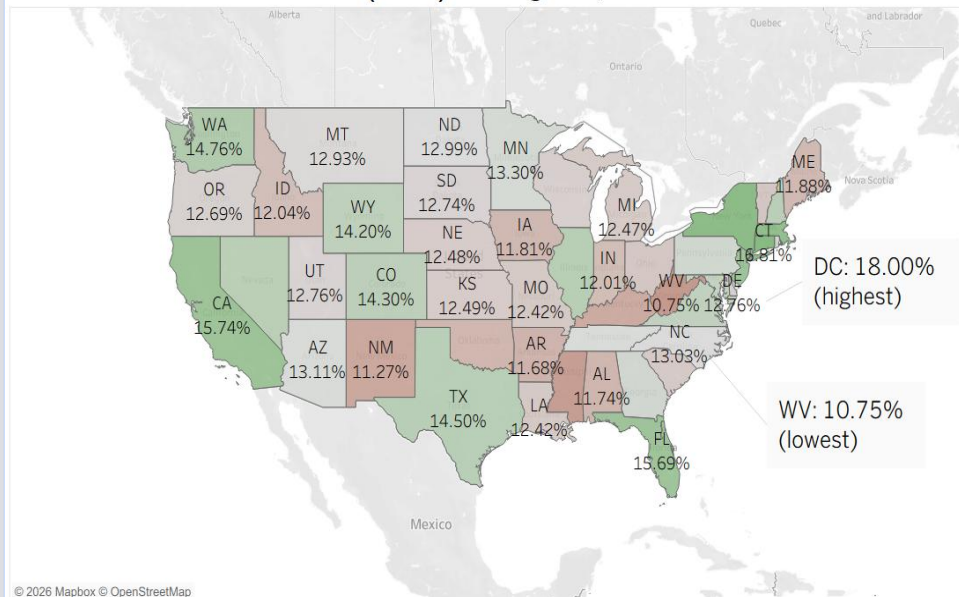
–1.32 percentage points

t = 5.58 | p < 0.0001 ✓ SIGNIFICANT

Chart shows top 5 brackets for clarity; statistics computed across all 10 brackets (significant in 9 of 10)

State-Level Variation: A Statistical Mirage

U.S. Choropleth Map — ETR by State



ETR Range

DC: 18.00% (highest)

WV: 10.75% (lowest)

ANOVA p-value

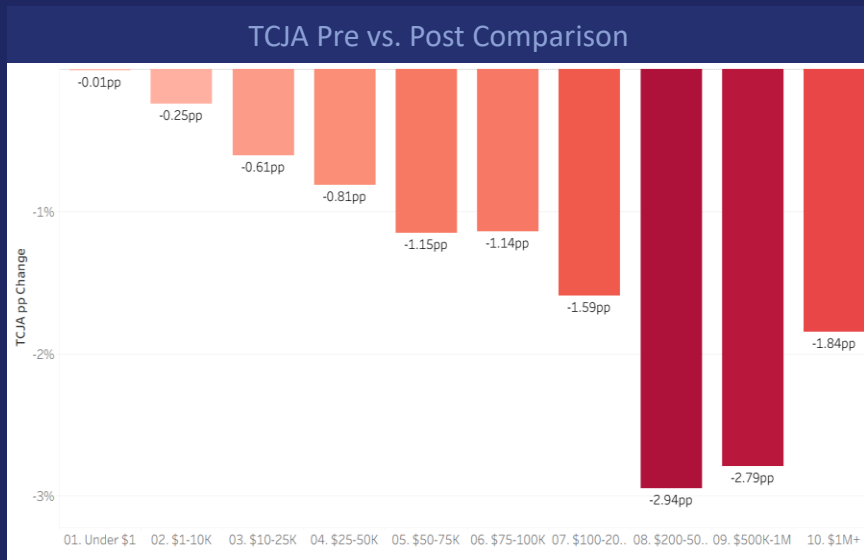
0.88

NOT statistically significant

Simpson's Paradox

Apparent state disparity is driven by **INCOME COMPOSITION**, not tax policy. Federal tax law is uniform.

Who Saw the Largest Changes?



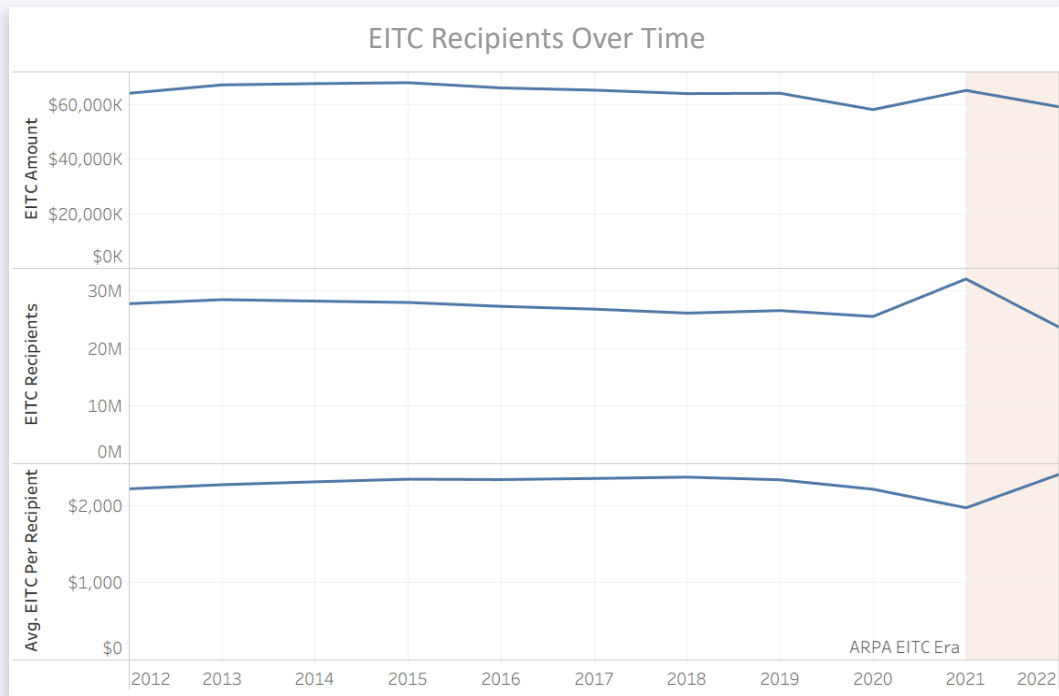
TCJA Effect by Bracket

\$50–75K	–1.15 pp
\$100–200K	–1.59 pp
\$200–500K	–2.94 pp LARGEST
\$500K–1M	–2.79 pp
\$1M+	–1.84 pp

Why Uneven?

- Capital gains rates (already lower) — unchanged
- SALT deduction cap — affects high-income states
- Standard deduction doubling — helps middle brackets

The ARPA EITC "Credit Cliff"



American Rescue Plan Act

2021

2020 **26.5M**

2021 **32.0M** **+5.5M ↑**

2022 **23.7M** **-8.4M ↓**

8.4M workers lost the credit in 2022

Key Takeaways from the Analysis

1

The U.S. tax system is statistically progressive

$r = 0.96$

2

TCJA reduced effective rates measurably

$-1.32 \text{ pp}, p < 0.0001$

3

Policy impact varies by income composition

Capital gains explains the surprise

4

Temporary credits cause volatility

ARPA $\rightarrow$ 8.4M credit cliff

Data analysis reveals nuance that headlines miss.

Thank You

Wonseok Lee

Virginia Tech × Fullstack Academy DA Bootcamp



Tableau Dashboard: public.tableau.com/app/profile/wonseok.lee8183/viz/CapstoneProjectIncomevsTaxBurden_WonseokLee/CapstoneDashboard



GitHub: github.com/wonseok3lee-spec/vt-fullstack-bootcamp



Connect: [linkedin.com/in/wonseok-lee-cpa/](https://www.linkedin.com/in/wonseok-lee-cpa/)



Data Source: IRS Statistics of Income — Historic Table 2 (Tax Years 2012–2022)
irs.gov/statistics/soi-tax-stats-historic-table-2

Questions?